PSM V1 — CONNECTION-LEVEL SCHEMATIC (thesis baseline)

16x VCNL4020 on 4x4 array; 2x PCA9548A; Tiva/LaunchPad header. PCB is authoritative for passives and routing.

Per sensor: 100nF local decoupling (C1–C16), 10R IR-anode series resistor (R1–R16), 22uF bulk C17/C18.

U17 channels 0–7 -> U1..U8; U18 channels 0–7 -> U9..U16.

VCNL4020: 1 IR anode, 2 SDA, 3 INT, 4 SCL, 5 VDD, 8/9 GND, 6/7/10 NC.

Carleton University — Instrumented Infant-Child Car Seat

Sheets: /

File: psm-v1.kicad_sch

Title: PSM V1 — Connection-Level Schematic

Size: A1 | Date: 2025-06-04 | Rev: V1.0

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